



## Overview

The **GPX-1™** is a multi-frequency and multi-constellation GNSS receiver module that combines GPS/QZSS (L1, L5), Galileo (E1, E5), GLONASS (L1), BeiDou (B1, B2), NavIC (L5), and SBAS (L1) to provide improved performance for global positioning navigation solutions. The use of multiple frequencies greatly reduces multi-path effects in urban environments and improves the location accuracy.

The **GPX-1-RTK™** has onboard Real-Time Kinematic (RTK), enabling centimeter level position accuracy with an RTCM3 correction input stream.

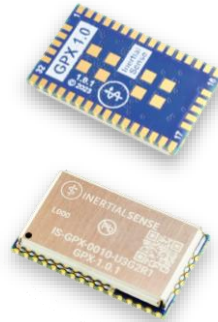
The **GPX-1-Dual™** has two GNSS receivers (two antenna channels) for RTK positioning and dual GNSS heading/compassing, eliminating the need for magnetometer heading.

The GPX-1 comes in a 20.7 x 12.5 mm LGA surface mount module and includes a powerful baseband processor, embedded Flash memory, and integrated LNA. The ultrasensitive RF front-end and multi-frequency and multi constellation capability support navigation in challenging outdoor scenarios.

Combining the GPX-1 and **IMX-5™** tactical grade IMU/INS creates GNSS aided inertial navigation sensor fusion with roll, pitch, heading, velocity, and position up to 250Hz.

The **RUG4-IMX5-GPX1™** combines the GPX-1 and IMX-5 tactical grade IMU/INS in a rugged aluminum enclosure and RS232, RS485, and CAN bus. Inertial navigation sensor fusion is enabled for roll, pitch, heading, velocity, and position.

The **Inertial Sense SDK** is an open-source software development kit for quick integration to configure and communicate with Inertial Sense products. The SDK includes data logger, math libraries, and interface for Linux, Windows, and embedded platforms.



**GPX-1**  
 Size: 20.7 x 12.5 x 2.9 mm  
 Weight: 1.7 g  
 GNSS: Multi-Band L1/L5



**RUG4-IMX5-GPX1**  
 Size: 30.5 x 25.4 x 10.5 mm  
 Weight: 14 g  
 GNSS-INS: Multi-Band L1/L5

## Features

- **Multi-band (L1/L5) GNSS receiver**
- **Multi-constellation (GPS, GLONASS, QZSS, BeiDou, Galileo)**
- **Dual GNSS receivers (two antennas)**
- **Onboard RTK Positioning and Compassing**
- **Low power consumption GNSS positioning**
- **Combine w/ IMX-5 for GNSS aided INS @ 250Hz**
- **0.4° RMS heading accuracy @ 1m baseline**
- Ultra-sensitive -165 dBm (tracking) RF front-end
- Supports ephemeris file injection (A-GNSS)
- Satellite Based Augmentation System (SBAS)
- Up to 10 Hz output data rate
- -40°C to 85°C Operating Temperature
- Binary and NMEA Protocol
- PPS Output for Time Synchronization
- SDK, Example Software, and Data Logging

## Applications

- Drone Navigation
- Unmanned Vehicle Payloads
- Ground and Aerial Survey
- Automotive Navigation
- Stabilized Platforms
- Antenna and Camera Pointing
- First Responder and Trackers
- Health, Fitness, and Sport Monitors
- Robotics, Ground Vehicles, Maritime

## Specifications

| Features - Performance                |                                                |                        |
|---------------------------------------|------------------------------------------------|------------------------|
| Heading accuracy                      | 0.4° RMS using 1m antenna baseline *           |                        |
| Position accuracy                     | 1.0 m CEP                                      | RTK: 0.02 m CEP **     |
| Vertical Operational Limit            | 100,000 m                                      |                        |
| Receiver type                         | 62 physical acquisition/tracking channels      |                        |
| Constellations                        | GPS (L1C/A L5)                                 | GLONASS (L1OF)         |
| (Frequency bands)                     | Galileo (E1B/C, E5a)                           | BeiDou (B1I, B1C, B2a) |
|                                       | QZSS (L1C/A, L1S, L1C/B, L5)                   | NavIC (L5)             |
|                                       | SBAS (L1): WAAS, EGNOS, MSAS, GAGAN            |                        |
| Navigation output rate                | Up to 10 Hz                                    | (Position & Velocity)  |
| Compassing output rate                | Up to 5 Hz                                     | (Heading)              |
| Convergence time                      | 1 s                                            | RTK: < 10 s            |
| Acquisition                           | Cold start                                     | 24 s                   |
|                                       | Hot start                                      | 1 s                    |
| Sensitivity                           | Cold start                                     | -149 dBm               |
|                                       | Hot start                                      | -158 dBm               |
|                                       | Reacquisition                                  | -163 dBm               |
|                                       | Tracking & nav.                                | -167 dBm               |
| Max Velocity                          | 600 m/s (airborne mode), 150 m/s (other modes) |                        |
| Internal LNA gain                     | 69 dB                                          |                        |
| 1 PPS Output                          | 10 ns resolution                               | < 100 ns accuracy      |
| Oscillator                            | TCXO                                           |                        |
| RTC crystal                           | Built-in                                       |                        |
| Anti-jamming                          | 7-ch notch filter for each L1 and L5 band      |                        |
| Memory                                | Flash                                          |                        |
| Moving base                           | For dual GNSS compassing (heading)             |                        |
| Supported antennas                    | Active                                         |                        |
| * GNSS-Compassing, ** RTK-Positioning |                                                |                        |

| Interfaces                 |                                      |  |  |
|----------------------------|--------------------------------------|--|--|
| Serial (GPX-1)             | UART x3, SPI, I2C, CAN, USB          |  |  |
| Serial (RUG-4)             | UART x2, SPI, RS232, RS485, CAN, USB |  |  |
| Max Baud Rate:             |                                      |  |  |
| SPI                        | 10 Mbps                              |  |  |
| UART, RS422, RS485         | 3 Mbps                               |  |  |
| RS232                      | 500 Kbps                             |  |  |
| I/O Level (UART, SPI, PPS) | 1.8V to 3.3V                         |  |  |

| Package                             |                              |            |
|-------------------------------------|------------------------------|------------|
| Package                             | 42-pin LGA (Land Grid Array) | SMT module |
| Size                                | 20.7 x 12.5 x 2.9 mm         |            |
| Weight                              | 1.7 g                        |            |
| Effective Thermal Resistance (Oeff) | 15                           | °C/W       |
| Thermal Resistance (ΘJA, ΘJB, ΘJC)  | 42.3, 27.5, 1.6              | °C/W       |

| Function                              | GPX-1™ | +RTK | +Dual | +IMX-5™ |
|---------------------------------------|--------|------|-------|---------|
| Position and Velocity                 | •      | •    | •     | •       |
| RTK Centimeter Level Position         |        | •    | •     | •       |
| Dual GNSS Compassing (Heading)        |        |      | •     | •       |
| Roll, Pitch, Velocity, Position (INS) |        |      |       | •       |

## Part Numbers

|                          |                                              |
|--------------------------|----------------------------------------------|
| IS-GPX-0010-U3G2-IND     | Dual GNSS (no compassing or RTK positioning) |
| IS-GPX-0010-U3G2-C1-IND  | GNSS-Compassing                              |
| IS-GPX-0010-U3G2-R1-IND  | RTK-positioning (available mid 2024)         |
| IS-IG-2050-U3G2-C1R1-DVK | Development Kit for IG2-IMX5-GPX1            |
| ANN-MB1-00               | Multi-band (L1/L5) active GNSS antenna       |



**Development kits available on our website.**

| Environmental                   | MAX                                |
|---------------------------------|------------------------------------|
| Operating Temperature           | -40 to 85 °C                       |
| Storage Temperature             | -40 to 85 °C                       |
| ESD rating                      | ± 2 kV Human body model            |
| Solder Reflow Temperature Max   | 245 °C                             |
| Solder Reflow Temperature Limit | 217 °C liquidus: 40 – 60 s         |
| Magnetic field immunity         | 25 mT (operation), 55 mT (storage) |

| Electrical                              |      |      |      |       |
|-----------------------------------------|------|------|------|-------|
|                                         | Min  | Typ  | Max  | Units |
| Power Draw @ 5Hz                        |      | 160* | 200* | mW    |
| Power Draw @ 25Hz                       |      | 190* | 240* | mW    |
| Supply Voltage (Vcc)                    | 3.0  | 3.3  | 3.6  | V     |
| I/O Pin MAX Voltage Range               | -0.5 |      | 3.6  | V     |
| Total Output Current, All Pins          |      |      | 100  | mA    |
| Logic levels for 3.3V I/O (VAUX = 3.3V) |      |      |      |       |
| Input low-level                         |      |      | 0.99 | V     |
| Input high-level                        | 2.31 | 3.3  |      | V     |
| Output high-level                       |      | 3.3  |      | V     |
| Logic levels for 1.8V I/O (VAUX = 1.8V) |      |      |      |       |
| Input low-level                         |      |      | 0.4  | V     |
| Input high-level                        | 1.3  | 1.8  |      | V     |
| Output high-level                       |      | 1.8  |      | V     |
| RF Power In (GNSS1_RF, GNSS2_RF)        |      |      | 0    | dBm   |
| VCC_RF Output Current** @3.0V           |      |      | 30   | mA    |
| Active L1/L5 Antenna Gain               | 15   |      | 30   | dB    |

\* Power draw does not include active antenna power draw.

\*\* VCC\_RF is supplied from VAUX through a load switch and a 10Q 0.2W resistor.

| Related Products: RUG-4 & IG-2 Electrical |     |      |     |       |
|-------------------------------------------|-----|------|-----|-------|
|                                           | Min | Typ  | Max | Units |
| Supply Voltage (VIN)                      | 4.5 |      | 20  | V     |
| RUG4-IMX5-GPX1 + Antenna                  |     |      |     |       |
| Current Draw @ 5V, 250Hz*                 |     | 185  |     | mA    |
| Power Consumption @250Hz*                 |     | 927  |     | mW    |
| Power Consumption @100Hz*                 |     |      |     | mW    |
| Power Consumption – Dual                  |     | 1470 |     | mW    |

\*Navigation filter update rate.

| Related Products: RUG-4 Package |                                              |  |  |
|---------------------------------|----------------------------------------------|--|--|
| Size                            | 30.5 x 25.4 x 10.5 mm                        |  |  |
| IP Rating                       | 40                                           |  |  |
| Mounting Tab                    | 30.836 mm                                    |  |  |
| Weight                          | 14.0 g                                       |  |  |
| Connectors                      | Main: Harwin# G125-MV11205L1P, GPS 1/2: MMCX |  |  |

| Related Products: IG-2 Package |                              |            |
|--------------------------------|------------------------------|------------|
| Package                        | 36-pin LGA (Land Grid Array) | SMT module |
| Size                           | 46.6 x 24.5 x 5.9 mm         |            |
| Weight                         | 8.5 g                        |            |



**IG-2**  
 SMT Module (GPX1 + IMX5)  
 Size: 46.6 x 24.5 x 5.9 mm  
 Weight: 8.5 g  
 GNSS-INS: Multi-Band L1/L5

